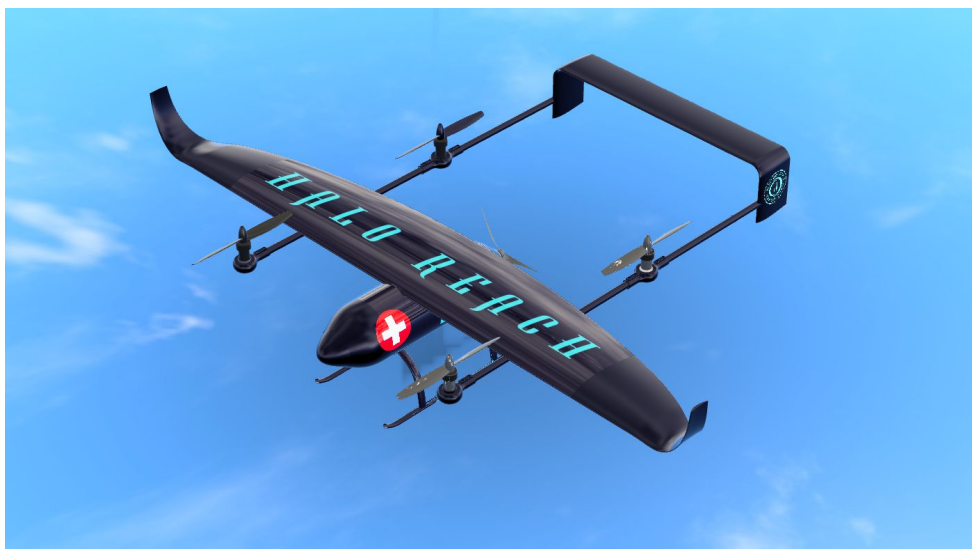

HALO REACH: Fuselage and Booms Structural Design

UAS for Disaster Relief - AE3 2024



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Date: 17/06/2024
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Abstract

This objective of the UAS Disaster Relief project was to design a versatile, modular Vertical Take-Off and Landing (VTOL) Uncrewed Aerial System (UAS) that is capable of performing vertical take-offs and landings from ship decks under high-wind conditions, cruise to and from disaster sites, and serve as a long-endurance communications relay or deliver relief supplies by landing vertically. This report details an intricate analysis of the loads subjected to the fuselage; establishing that the most constraining load case is landing at MTOW during the supply mission. This played a critical role in sizing the internal structure. This report also encompasses an outline of the design process, and the justification for using composites; an optimisation of the composite layup and internal structure has also been detailed. Finally, the structural sizing and design of the booms has also been disclosed. FEA Analysis has been used to conclude the preliminary design of the fuselage and the booms, verifying the designs.

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1 Introduction

In the wake of a natural disaster, it is often the case that conventional routes for ground vehicles and fixed-wing aircraft become inaccessible due to as road damage, flooding, or runway destruction. This has motivated the development of HALO Reach’s highly versatile VTOL Uncrewed Aerial System. As required by the project brief, the UAS has been designed to complete critical tasks including: vertical take-off and landing, cruise between disaster sites, communications relay and supplies delivery.

To satisfy the requirements of both long-endurance and supplies-delivery missions, the vehicle has been modified using a system of modular add-ons, which required the development of a fuselage that could support this. This report describes the conceptual preliminary airframe design of such the UAS, optimized for rapid deployment and versatility in disaster relief scenarios. Following the completion of the conceptual design stage which focused on the initial sizing and analysis of the primary structural parts of the fuselage and booms; the preliminary airframe design was conducted. This process included evaluating specific load cases that the UAS would experience during different stages of its mission:

- **Load Case 1:** Cruise at MTOW, ETOW and OEI for supply and endurance missions.
- **Load Case 3:** Landing nose-off at MTOW, ETOW and OEI for supply and endurance missions.
- **Load Case 4:** Take-off at MTOW, ETOW and OEI for supply and endurance missions.

Each load case is evaluated for the fuselage and booms, with the final design being governed by the most constraining load case for each particular structure; load analysis of the fuselage required load analysis of the wing, tailplane and booms first [1]. A safety factor of 1.5 has been applied where appropriate and manufacturing constraints have also been considered.

The design of aircraft structures must strike a balance between providing sufficient strength to withstand critical load cases whilst also being structurally efficient, lightweight, and reliable. This report will highlight the various design processes and decisions made, supported by assumptions and design trade-offs where necessary, demonstrating how an efficient iteration of the airframe was achieved.

The details of procedures that aid the design of the airframe, namely the selection of aluminum alloys and composite materials for the fuselage, has been highlighted in Table 2 [2], and the materials selected for the sandwich structure of the booms have also been determined in Table 3 [3].

Table 2. Material Selection and Properties of the Fuselage and Boom Skin Panels

Material	Components	Tensile Strength	Compressive Strength	Shear Strength	Density
Toray Prepreg 3960	Skin,Stringers, Longerons, Light-frames, Boom Skin	3759 MPa	338 MPa	135 MPa	1570 kg/m^3
Al 2024-T4	Heavy-Frames	395 MPa	400 MPa	124 MPa	2767 kg/m^3

Table 3. Material Selection of Sandwich Structure Core

Material	Part	Strength in L-Direction	Strength in W-Direction	Compressive Strength	Density
A1-Nomex Honeycomb	Core	3.6 MPa	2.2 MPa	23 MPa	200 kg/m^3

A detailed sketch of the final design of the UAS has been provided in the Appendix (A), CAD models were provided and modified upon the request of all teams as adjustments were made in the design of the UAS throughout the conceptual and preliminary design stages.

2 Fuselage

2.1 Initial Sizing of the Fuselage

The initial sizing of the fuselage structure required that the fuselage may be able to house the sub-systems and modular add-ons that are required to complete the mission as described by the mission brief. The sizing process has been outlined in Figure 2, and Figure 3 displays how the fuselage was sized to house all internal components on Fusion.

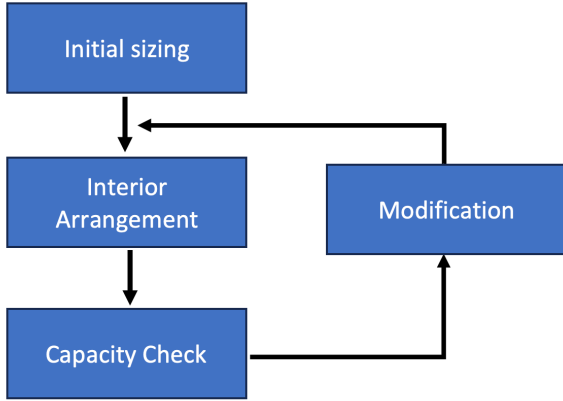


Figure 2. Initial Sizing Process of the Fuselage

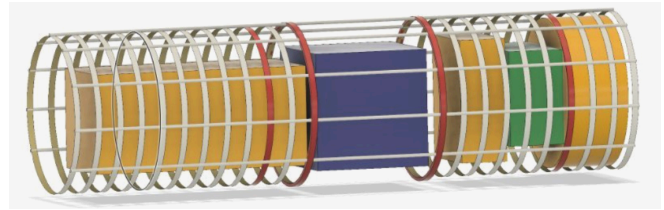


Figure 3. Sizing the Fuselage on Fusion

Following this procedure the following dimensions were imposed for the fuselage geometry:

Table 4. Final Fuselage Dimensions

Fuselage Length(m)	Diameter(m)	Nose Length(m)	Tail Length(m)
1.9	0.55	0.2	0.25

2.2 Load Analysis

The preliminary design of the fuselage began with the analysis of the three load cases: cruise, landing and take-off. The fuselage was modelled as a constant diameter cylindrical section, where the nose and empennage fuselage geometries were neglected from this preliminary analysis along with the effects of cutouts and secondary structures. As the brief required that the aircraft must perform a supply drop and a communications relay, the most constraining cases were evaluated for each load case; this required considering variations of payload and fuel weights for both the supply drop mission and long endurance mission, for each load case the impact of OEI was also determined.

The inertial load distribution for the supply mission at MTOW was first determined as this mission appeared to be the most constraining, this was used evaluate the three load cases; the inertial load distribution consisted of a combination of distributed masses along the length of the fuselage multiplied by the gravitational constant (g), the load distribution obtained at MTOW has been evaluated in Table 4. With the load distribution known for a load factor of 1, these distributions were scaled to reflect different load factors at each critical flight phase. As seen in Figure 5 the load is non-zero along the fuselage due to the contribution of the distributed fuselage weight and the subsystems.

For cruise, aerodynamic loads and inertial loads were considered along with the thrust generated by the two rotors on each boom. The aerodynamic forces and the reaction forces generated by the rotors on the aircraft were represented by loads on the front and rear wing spars (F_{FS} and F_{RS} respectively) to balance the aircraft's weight and wing torque. With the reaction forces on the fuselage known, the inertial load distribution was superposed with the aerodynamic load distribution and reaction forces to produce a total load distribution. A shear force distribution was then produced and subsequently

Component	Mass (kg)	X_{CG} (m)
Payload	50	1.1
Engines	11.5	1.79
Fuel	6.5	0.76
Fuel systems	2	1.6
Batteries	5.4	0.31
Satellite	10.2	0.38
Wings	16.72	1.15
Controls	10.2	0.19
Pusher	0.56	1.9
Landing Gear	3.5	1.1
Add-ons	7	1.1

Figure 4. Weight Distribution at MTOW

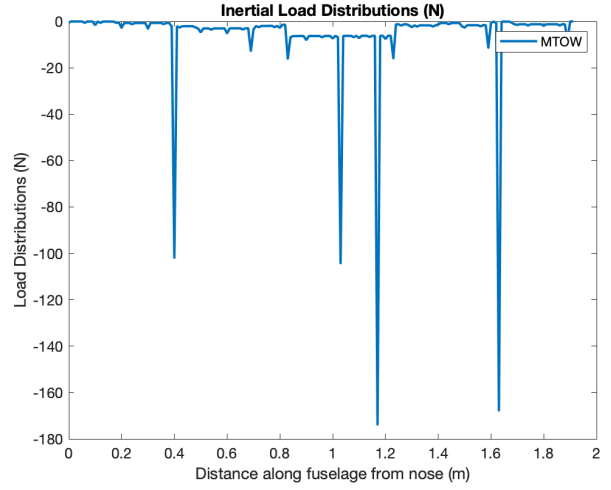


Figure 5. Inertial Load distribution

integrated using Equation 1 and the `cumtrapz()` function from MATLAB to yield the final bending moment distribution.

$$M(x) = \int S(x) dx \quad (1)$$

The aircraft was modelled as a simple beam with free ends. This imposed the constraint that the bending moment at the tail had to be zero. Therefore, an additional vertical force from the horizontal stabiliser (F_{HS}) had to be considered to enforce this boundary condition. To find the load that enforces this boundary condition, the problem was cast in matrix form and iterated through in MATLAB to find the load that sets the bending moment to zero with a tolerance of 1Nm. The matrix form is as below in Equation 2, using the distances between each wing spar and CG, (x_{FS} and x_{RS}), F_z refers to inertial loads.

$$\begin{bmatrix} x_{FS} & x_{RS} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} F_{FS} \\ F_{RS} \end{bmatrix} = \begin{bmatrix} \Sigma M \\ \Sigma F_{HS} + F_z \end{bmatrix} \quad (2)$$

The next load case evaluated was a nose-off landing where aerodynamic loads from the wings were neglected. In this case, the full weight of the aircraft was supported by a reaction force from the main landing gear (F_{MLG}) and the reaction force of the rotors (F_R), both of which appeared to act at the locations of the front and rear spars of the wing due to the design of the aircraft. A vertical force from the horizontal stabiliser also acted to satisfy the tail bending moment constraint. The iterative matrix problem for the landing case can be seen in Equation 3.

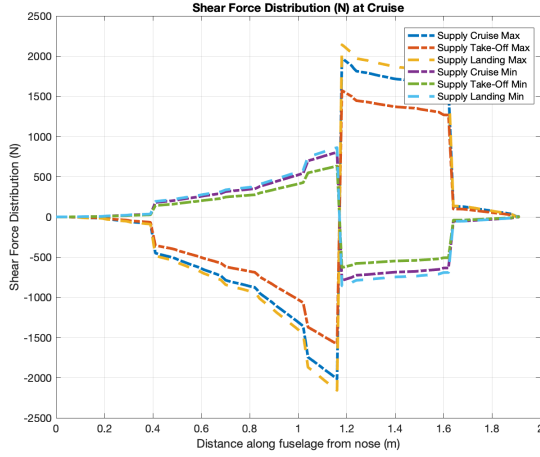
$$\begin{bmatrix} x_{FS} & x_{RS} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} F_{MLG} + F_R \\ F_{HS} \end{bmatrix} = \begin{bmatrix} \Sigma M \\ \Sigma F_z \end{bmatrix} \quad (3)$$

The final load case evaluated was at take-off. In this case, the full weight of the aircraft was supported by a reaction force generated by the thrust of the rotors (F_R), both of which appeared to act at the locations of the front and rear spars of the wing. A vertical force from the horizontal stabiliser also acted to satisfy the bending moment constraint. This iterative matrix problem for the take-off can be seen in Equation 4.

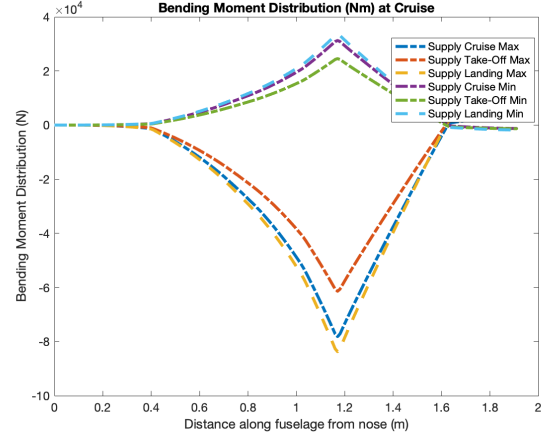
$$\begin{bmatrix} x_{FS} & x_{RS} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} F_R \\ F_{HS} \end{bmatrix} = \begin{bmatrix} \Sigma M \\ \Sigma F_z \end{bmatrix} \quad (4)$$

Comparisons between the shear and bending moment distributions of all three load cases are displayed in Figures 6a and 6b at the maximum load factor of 3.75 and minimum load factor of -1.5. Although differing in magnitude, the distributions did not vary significantly because the position of the landing gear lay close to the wing spars, thus the loads on the wing spars were similar to the main landing gear

when normalised by load factor. As expected, the landing load case had the highest shear force and bending moment distributions. Therefore, this was the critical load case used for fuselage sizing.



(a) Shear Force Distributions



(b) Bending Moment Distributions

2.3 Fuselage Structural Design

The primary structural component considered was the cylindrical skin which resisted shear forces and encased the longitudinal stringers running the entire length of the aircraft from nose to tail [4]. To withstand buckling loads, circular frames were placed along the length of the aircraft. The majority of these frames were classified as light frames whose purpose was to withstand the distributed loads along the aircraft [5]. In comparison, heavy frames were placed at points of high load concentrations such as the wing spar positions and payload frame positions. The interaction between components within the fuselage was complex thus it was deemed appropriate to make some idealisations. The first was to assume that the fuselage was cylindrical along its entire length, thus taper at the nose and empennage was neglected. The second allowed for the skin and stringers to be modelled using boom shear panel idealisation; the stringers and skin were modelled as booms and shear panels respectively [6].

2.4 Composite Layup Design

2.4.1 Classic Laminate Theory

The composite layup design utilized classical laminate theory to evaluate laminate performance [7]. Key properties were determined using empirical relations and the rule of mixtures. The Young's modulus of the composite in fiber and transverse directions was calculated using Equations 5 and 6:

$$E_1 = E_f v_f + E_m v_m \quad (5)$$

$$E_2 = \left(\frac{v_f}{E_f} + \frac{v_m}{E_m} \right)^{-1} \quad (6)$$

Composite strength in both directions was determined by Equations 7 and 8:

$$\sigma_1 = \sigma_f v_f + \sigma_m v_m \quad (7)$$

$$\sigma_2 = \sigma \left(1 - 2 \sqrt{\frac{v_f}{\pi}} \right) \quad (8)$$

Using the rotation matrix, local stresses for a particular fiber orientation were derived from global stresses by Equation 9:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_6 \end{bmatrix} = [T_\sigma] \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix} \quad (9)$$

The laminate's response was determined using the ABD matrix (Equation 10):

$$\begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{bmatrix} \epsilon^0 \\ \kappa \end{bmatrix} \quad (10)$$

2.4.2 Laminate Layup Criteria

The composite design of structural members required consideration of additional failure mechanisms to yield and fatigue, such as delamination of laminate layers. The first ply failure of a composite may lead to significant decline in structural performance as well as progressive failure of laminae until ultimate failure. Deficiencies in the transverse strength properties of composite structures makes delamination a critical failure mechanism [8]. As both the fuselage and the booms have been modeled as cylindrical sections, the number of plies along the span will be kept constant. The possibility of in-plane failure as a consequence of delamination from ply-drops is thus avoided.

Ply grouping reduces damage resistance, hence this has been avoided, and for laminates with a 45° interface angle, the first ply must be oriented along the shortest dimension. Placing multiple plies of the same orientation together should be avoided, but if unavoidable, stacking has been limited. To reduce the possibility of delamination, a balanced and symmetric layup about the midplane is essential. Additionally, by using only 0° , 90° and $\pm 45^\circ$ plies, the layup is quasi-isotropic. Quasi-isotropic layups have greater damage resistance, which is crucial considering the mission requirements of the UAS. Following guidelines, the layup design criteria has been summarised below [9]:

- place an even number of $\pm 45^\circ$ plies about the mid-plane
- avoid placing 0° plies at the outer plies
- avoid interface angles of 90°
- place a maximum of 2 plies of the same orientation in a row

2.4.3 Failure Criteria and Buckling Analysis

In dealing with composite materials, it was essential to consider failure due to their complex and anisotropic nature. Analysis has been conducted at first ply failure and the failure criteria used was the Tsai- Wu failure criteria. The Tsai-Wu failure criterion is an interactive, non-linear, stress-based criterion; it's only limitation is that it does not predict the failure mode [10]. Failure is predicted under plane state of stress as seen in Equation 11 :

$$F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2 = 1 \quad (11)$$

The coefficients of F are defined in terms of known strengths of the composite material: X , Y and S ; the subscripts refer to the tensile, compressive and shear strengths in different directions :

$$F_1 = \frac{1}{X_t} - \frac{1}{X_c}, \quad F_2 = \frac{1}{Y_t} - \frac{1}{Y_c}, \quad F_{11} = \frac{1}{X_t X_c}, \quad F_{22} = \frac{1}{Y_t Y_c}, \quad F_{66} = \frac{1}{S^2}, \quad F_{12} = -\frac{1}{2} \sqrt{F_{11} F_{22}}$$

Additionally, as a result of bending moments imposed on the structural components, direct buckling is another mode of failure to be considered, Equation 12 was used to compute the direct critical buckling load.

$$N_x^{cr} = \frac{2\pi^2}{w^2} (\sqrt{D_{11} D_{22}} + (D_{12} - D_{66})) \quad (12)$$

Alongside direct buckling, another potential failure mode that was considered was shear buckling, this was determined using Equation 13.

$$N_{xy}^{cr} = \frac{4}{w^2} (\sqrt{D_{11} D_{22}^3} \times (8.125 + 5.045K)), \quad K = \frac{2D_{66} + D_{12}}{\sqrt{D_{11} D_{22}}} \quad (13)$$

The buckling criteria was a combination of direct and shear buckling summarised by Equation 14.

$$\frac{N_{xy}^2}{N_{xy}^{cr}} + \frac{N_x}{N_x^{cr}} \leq 0.99 \quad (14)$$

The final laminate stacking sequence would be checked against the failure criteria and buckling analysis to optimise the layup as to avoid failure.

2.4.4 Overall Procedure For Composite Layup Design

A MATLAB code has been formulated such that a systematic and iterative procedure is conducted to determine optimal composite layups for the skin, stringers light frames and longerons of the fuselage. The process for each structural component began with initializing parameters such as the number of plies and the possible ply orientations (0° , $\pm 45^\circ$, 90°), it was found that the minimum number of plies to withstand loading was 8.

The code then generated all possible symmetric layup combinations and applies a range of filters to ensure that the layups meet the layup criteria; this included ensuring the layup complied with the 10/70 rule, making sure combinations with more than two consecutive plies of the same orientation were removed and checking the equality of $\pm 45^\circ$ ply proportions to ensure balanced shear properties. The code then calculated the laminate properties (such as Young's Modulus, Shear modulus, and Poisson's ratio) by computing the ABD matrix for each valid layup. All anisotropic layups are eliminated, allowing for uniform material properties.

For the stringer light frame layup and longeron, the process was similar, however, additional steps were taken to evaluate different stringer geometries. Potential stringer geometries were assessed for each combination of layup and geometry based on critical buckling stress and the Tsai-Wu failure criteria. Layups that did not meet the structural and layup requirements were filtered out. The spar flange design involved calculating the required flange width and buckling loads, adding extra plies if necessary, and ensuring the final configurations met all performance criteria. This iterative procedure ensured all layups followed the Layup Design Criteria and was optimized for strength and efficiency.

2.5 Skin Stringer Design

The stringers run from the nose to the tail of the aircraft and have been designed to carry direct stresses, stiffen the fuselage, and prevent buckling. The skin of the fuselage is assumed to carry shear and pressurization loads, whilst the stringers are assumed to resist direct stresses.

The minimum allowable thickness of 1 mm due to manufacturing constraints hence these physical limits did not influence the skin thickness. However the design of the undercarriage made it so that the skin thickness was constrained to 2.2 mm. Following the composite design procedure, an optimal layup was found based on the layup and structural constraints. To further validate the composite skin design, the shear and pressurization loads were examined on the fuselage skin, as under maximum loading the fuselage cannot fail [11]. The total shear flow (q_{total}) in a skin section has been calculated using Equation 15, with the radius (r), tangential load (P), radial load (Q), angle from the bottom of the fuselage (Φ), and angle from the bottom of the fuselage where the load is applied (β).

$$q_{\text{total}} = \frac{T + Pr}{2\pi r^2} + \frac{P \cos(\Phi - \beta)}{\pi r} + \frac{Q \sin(\Phi - \beta)}{\pi r} \quad (15)$$

The landing load case at the MTOW produces the highest q_{total} , thus this value has been used to determine the minimum fuselage skin thickness due to shear flow ($t_{\text{min, shear}}$) using Equation 16.

$$t_{\text{min, shear}} = \frac{q_{\text{total, max}}}{\sigma_y} = 0.07 \text{ mm} \quad (16)$$

With the minimum fuselage skin thickness known due to shear loads, the cabin pressurization loads were next to be analysed. The cabin would be pressurized to the atmospheric pressure at 500 m. The stresses acting on the fuselage skin as a result of the pressure difference are the longitudinal stress (σ_L)

and the hoop stress (σ_H). Equations 17 and 18 were used to find the minimum fuselage skin thickness by evaluating the respective stresses against the tensile strength of the fuselage skin material.

$$\sigma_H = \frac{Pd}{2t} \quad (17)$$

$$\sigma_L = \frac{Pd}{4t} \quad (18)$$

$$\sigma_S = \frac{Pd}{4t} \quad (19)$$

Another requirement of an adequately designed fuselage is the minimization of bending moments resulting from movement of components within the fuselage [11]. To find the minimum thickness requirement, the cylindrical fuselage assumption was then amended by the addition of spherical caps. The spherical cap thicknesses were found from Equation 19. The minimum fuselage skin thickness criterion was evaluated using the Poisson's ratio of the skin for its particular layup, where $t_{\min, \text{cabin}} = 0.06$ mm. In combining strains, Equation 20 was then derived, relating cylindrical skin thickness to hemispherical cap thickness.

$$\frac{t_1}{t_2} = \frac{2 - \nu}{1 - \nu} \quad (20)$$

Given $t_{\min, \text{shear}} = 0.06$ mm and $t_{\min, \text{cabin}} = 0.07$ mm, the minimum fuselage thickness from physical constraints yields a minimum fuselage skin thickness of 0.07 mm. This did not seem to constrain the skin thickness, hence a skin thickness of 2.2 mm was determined to be substantial.

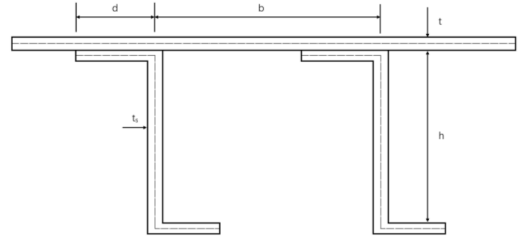
Table 5. Skin and Stringer Layup

Structural Member	Layup	Buckling Index	Tsai-Wu Index
Skin	[90/45/-45/0/-45/45]s	0.62	0.80
Stringers	[90/45/-45/0/0/-45/45]s	0.60	0.81

Table 6. Stringer Geometry in (mm)

t_{str}	2.56
d_{str}	11.0
h_{str}	27.5
b_{str}	300.0
Total Stringers	20

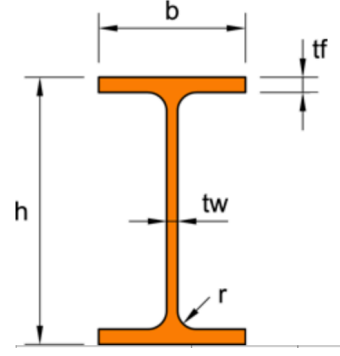
Figure 7. Cross-section with stringers



Following the design of typical aerospace skin-stringers, Z longitudinal stringers were adopted for their structural efficiency [11]. Z-stringers are defined by the parameters: thickness (t_{str}), height (h_{str}), flange length (d_{str}), and stringer spacing (b_{str}). Following the composite design procedure a layup for the stringers was also decided on as seen in Table 5. The optimal geometry for the stringers was then determined using the MATLAB `fmincon()` function [12], this took the largest and smallest values of each dimension produced by the the layup optimisation, and returned a geometry that minimised mass for that layup, the final geometry is shown in Table 6.

2.6 Longerons Design

Longerons are typically used in conjunction with frames and stringers and help distribute loads more evenly across the structure. This includes loads from the wings, landing gear, and internal components, this was essential to consider as the UAS features a retractable payload slot, thus the fuselage structure would require additional support. The longeron is designed to be an I-beam due to its' significant stiffness and strength properties, which given the operational environment, is a vital consideration as the mission may involve rough handling [13]. The design of the longeron involved the assumption that the web and flanges would be co-cured, and followed the composite layup procedure; this produced layups for both the web and flanges of the longeron, as seen in Table 8.

Figure 8. I-beam cross-section**Table 7.** Longerons web and Flange in (mm)

t_w	2.2
h_w	20.0
t_f	2.52
b_f	18.3
Total Longerons	2

Table 8. Longerons Web and Flange Layup

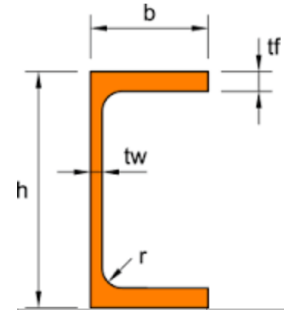
Structural Member	Layup	Buckling Index	Tsai-Wu Index
Web	[90/45/-45/0/-45/45]s	0.62	0.79
Flange	[90/45/-45/0/0/-45/45]s	0.60	0.81

Using the MATLAB `fmincon()` function a range of geometries were sorted through such that the dimension constraints were met, whilst also minimising mass for the particular layups found. The final geometry is shown in Table 7.

2.7 Light Frame Design

The light frames maintain the fuselage cross-section and distribute the shear loads and stress at their particular locations, they also provide panel stability for the overall strength and rigidity of the fuselage structure [11]. The cross-section profile chosen for the light frames was a C-shaped section; this was chosen as the design was constrained by the Z-shaped stringers that would pass through the light frames. Following the composite layup design procedure, an optimal layup was found, this is shown in Table 8.

Much like the stringers, the optimal geometry was determined using the MATLAB `fmincon()` function and this returned a geometry that minimised mass for a given layup, the final geometry is in Table 9.

Figure 9. C-frame cross-section**Table 9.** Web and Flange in (mm)

t_w	1.46
h_w	20.0
t_f	1.46
b_f	10.0
Total Frames	10

Structural Member	Layup	Buckling Index	Tsai-Wu Index
C-Frames	[90/45/-45/0]s	0.85	0.93

2.8 Heavy Frame Design

The location of the front and rear wing spars joining the fuselage, as well as locations where the payload retract from the fuselage are subjected to high concentrations of stress. Thus, heavy frames are used at these four locations along the fuselage.

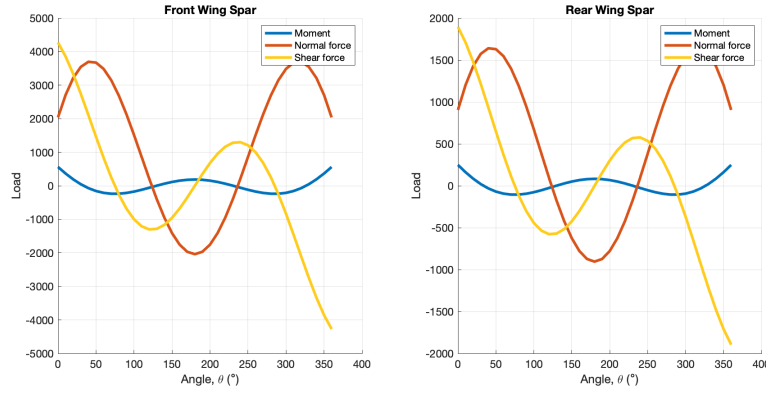


Figure 10. WISE Curves for the front and rear spars

Table 10. C-Frame Geometry in (mm) and Locations

Location	t_{lf}	h_{lf}	b_{lf}	Mass (kg)
Front Spar	3	25	10	0.6215
Payload Frame	5	25	10	0.7398
Payload Frame	5	25	10	0.7398
Rear Spar	3	25	10	0.6215

The heavy frames were sized based on the WISE equations [11], these equations have been detailed in the Appendix (B). Resolving each load into components and superposing the solutions, gave values for direct, shear and bending stress. These stresses varied radially around each frame, as seen in Figure 10, therefore the frames were sized based on the maximum stress values. The heavy frame thickness, base length and height were optimised such that the maximum bending and direct stresses from the heavy frame WISE curves were as close to the tensile yield stress of Al2024 as possible. The maximum shear stress was also optimised to be smaller than but as close to the shear yield stress of Al2024 as possible. A C-section cross section was chosen for the heavy frame design for ease of assembly [13].

2.9 Overview of Fuselage Structural Components

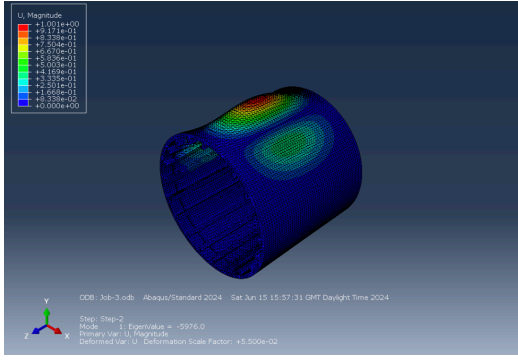
The overall breakdown for the component weight has been displayed in Table 11, it is evident that the objective of the structural optimisation has been met; from an initial mass estimation of 15.5 kg, the optimisation has reduced the mass by almost 3%, with a final mass of 15.1 kg.

Table 11. Weight of Structural Members

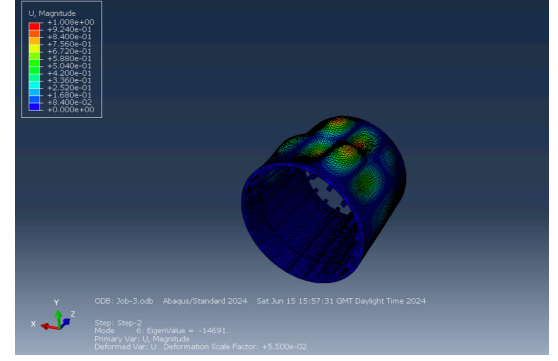
Structural Member	Mass (kg)
Skin	4.35
Stringers	2.74
Longerons	2.10
Light Frames	3.18
Heavy Frames	2.72

2.10 Verification

The buckling load is a critical consideration in the design of fuselage components, this is to ensure that the structure can withstand axial compressive loads without experiencing sudden failure. Verification of the fuselage structure required modelling the fuselage with the designed internal structure in Fusion; this was then imported into ABAQUS where loads and boundary conditions were applied. These loads corresponded to the most constraining loading case the fuselage undergoes- landing at MTOW.



(a) Primary Buckling Mode



(b) Secondary Buckling Mode

Figure 11. FEA Results

A composite material was incorporated on the model, and a layup was applied on each structural member to provide accurate prediction of the structural behavior. Buckling analysis was then performed to determine the critical buckling load. The buckling load was found to be 9735 N under the given loading. The primary buckling mode is shown in Figure 11a, the deformation pattern indicated global buckling. Higher-order modes were also identified, as shown in Figure 11b, providing insights into potential secondary buckling patterns. The primary buckling mode seems to be characterized by global buckling, whereas higher-order modes, indicate at both local and global buckling.

Table 12. Sensitivity Analysis

Mesh Size (m)	Buckling Load (N)	Displacement (mm)
0.01	14499	9.61
0.005	9735	9.49
0.003	5990	9.36
0.001	5976	9.17

A sensitivity analysis was conducted by varying the mesh size, the results show that the buckling load is highly sensitive to the mesh size, indicating that a finer mesh sizing is crucial for accurate analysis, Table 12 displays the results. The buckling load analysis reveals that the current fuselage design with the specified layup can withstand loads during landing, as the buckling load is never exceeded - the maximum load that the fuselage experience is 4388 N; hence it can be concluded that the design of the fuselage has been successful in preventing buckling.

3 Booms

3.1 Initial Sizing of the Booms

Length	3.45
Diameter	0.03
Thickness	0.01
Rotor 1	0.0
Rotor 2	2.15
Tail	3.01

Table 13. Boom Sizing in (m)

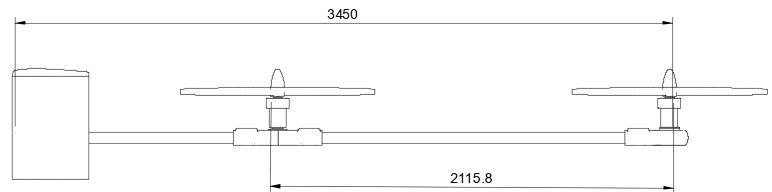


Figure 12. Rotor and Tailplane Positioning

The sizing of the booms was heavily interlinked with flight dynamics and hence the positioning and the sizing were constrained to satisfy stability requirements that the flight dynamics team imposed

[14]. The trade-offs between flight stability and structural integrity were then considered for the overall sizing such that the booms could provide sufficient support to the rotors and the tailplane. Table 13 summarises the sizing of the booms, whilst Figure 12 displays the positioning of the rotors and tailplane on the boom; the initial mass estimation for each boom was 2 kg, a design optimisation of this structure was then conducted to minimise mass.

3.2 Sandwich Structure Design

To evaluate the structural requirements of the boom, load analysis was first conducted [15]; the structural design corresponds to the most constraining load case to facilitate the design of the booms. Typically boom geometries are chosen accordingly to the mission profile of the aircraft, for the purpose of disaster relief, it was essential that the booms provided significant structural support to the rotors and the tailplane whilst also being light-weight. Hence a tubular geometry was chosen; considering the rotors were attached to the booms, a sandwich structure was deemed to be appropriate for these structural members due to their significant dampening qualities. To simplify the manufacturing of the booms, the member was designed using a constant thickness Nomex honeycomb core with and carbon fibre laminate face plates with a constant laminate layup, with material properties listed in Section 1.

The core thickness was then determined by iterating through thickness until axial stress in the skin was well below the in-plane strength of the face panel, this would be the point at which there would be failure due to yielding. The maximum stress is given by Equation 21, whilst the in-plane strength of the face is calculated for a layup [16], [17].

$$\sigma_{fx} = \frac{E_{fx}}{\sqrt{3(1 - \nu_{fx}^2)}} \frac{t}{R} \quad (21)$$

Following the composite layup procedure, a layup of [90/0/-45/45]s has been used for the face of the sandwich structure, this layup has an in-plane strength of 338 MPa. Iterating through this equation with a combination of values for d , t and c a final geometry was found, as shown in Table 14.

Structure	Thickness (mm)	Mass (kg)
Face Panels	1.46	1.26
Core	6.54	0.40
Total	9.46	1.66

Table 14. Geometry and mass of the structure

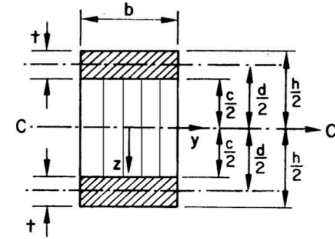


Figure 13. Reference Geometry [17]

The total mass of the boom was reduced by 17 %, hence the the mass savings achieved by using the Nomex core justify the higher associated cost. Additionally, the strategic orientation of fibers in the layup and the use of sandwich structures significantly minimized hoop stresses; thus, hoop stresses contribute minimally to the overall stress state and was neglected in analysis [18].

3.3 Skin Failure

Furthermore, to verify this design choice, two additional failure modes specific to the sandwich face plates were analysed. The first was face wrinkling, this is a buckling mode of the skin with a wavelength greater than the cell width, and buckling can occur in either direction depending on stiffness. The critical face wrinkling stress (σ_{fw}) was determined using Equation 22, where E_{fp} and E_c are the face plate and core Young's moduli, t_{fp} is the face plate thickness, and h_{sw} is the total sandwich thickness [17]. As the face wrinkling stress, calculated to be 4383 MPa, was greater than the maximum stress of 2081 MPa, it was determined the skin would not fail due to face wrinkling.

$$\sigma_{fw} = \left(\frac{2}{3} E_{fx} E_c \frac{t_{fp}}{h_{sw}} \right)^{1/2} \quad (22)$$

The second face plate failure mode evaluated was intra-cell dimpling, this occurs when the face plate becomes unsupported by the honeycomb walls, hence leading to buckling. The intra-cell buckling stress (σ_{fi}) of 3759 MPa was calculated using Equation 23 [17]. As this stress was significantly higher than the maximum stress within the face plate, the sandwich would not undergo intra-cell dimpling.

$$\sigma_{fi} = \frac{2E_{fx}}{1 - v_{fxy}^2} \left(\frac{2t_{fp}}{\alpha} \right)^2 \quad \alpha = 2t \sqrt{\frac{2}{1 - v_{fxy}^2} \frac{E_{fx}}{\sigma_{fY}}} \quad (23)$$

3.4 Core Failure

Core shear is a pertinent failure mode, and low density Nomex cores are particularly susceptible to this kind of failure. assuming brittle behaviour, failure is approximated to occur when mean shear stress of the core, given by Equation 24 exceeds the shear strength of the honeycomb structure. The shear strength of the honeycomb structure can be approximated by Equation 25 [17].

$$\tau_{cxz} = \frac{W}{2d} \quad (24) \quad \frac{\tau_{cs}}{E_{fx}} = 1.7 \left(\frac{\rho_c}{\rho_s} \right)^3 \quad (25)$$

Evaluating these equations it was found that this was core failure was not a concern as the shear stress of the honeycomb, calculated to be 615 MPa, was not exceeded.

3.5 Verification

Following a similar procedure to that of the fuselage, the boom was modelled on Fusion, then imported onto ABAQUS where structural analysis was conducted. To start with, loading was applied to the boom, this corresponded to the most constraining loading case that was determined in the load analysis of the booms [15]. Then boundary conditions were enforced such that the boom was modelled to be a cantilever beam. Following this, a layup was defined for the skin panels of the boom and material properties were also defined. Table 15 display the results of the FEA analysis, supporting FEA results have been provided in the Appendix (C).

Table 15. FEA Results

Mesh Size (m)	Buckling Load (N)	Displacement (mm)
0.01	4285 N	1.04

The results of the FEA signify that the boom has been designed such that buckling will not be a concern, this is simply due to the fact that the buckling load is never exceeded, and displacements due to the applied loading appear to be minimal such that they do not significantly influence stability.

4 Conclusions

The overall design of the UAS has been successful in meeting the mission requirements; the mass budget has been met whilst maximising structural integrity, this is supported by the FEA results. However, there are several areas for improvement; for starters, performing dynamic analysis would have enhanced the understanding and robustness of the design of the booms and fuselage. Dynamic analysis assesses the impact of time-dependent loads and vibrations; this would have allowed for targeted design modifications, improving the structure's performance under various loading scenarios- this is something to be considered for future iterations.

Furthermore, other failure modes such as fatigue and impact loading could have been assessed; and issues regarding cut-outs were overlooked for the simplification of the design, an analysis of the assembly of the UAS would have been a suitable progression from this design. As a result of the design freeze that the structures team had imposed in the final week, further refinements to the design regarding sizing could not be applied, but these changes would not have been drastic. Given the time availability, the overall structural design of both the fuselage and the booms have been adequate in meeting the structural requirements for both missions.

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Appendix

A CAD Sketch of the UAS

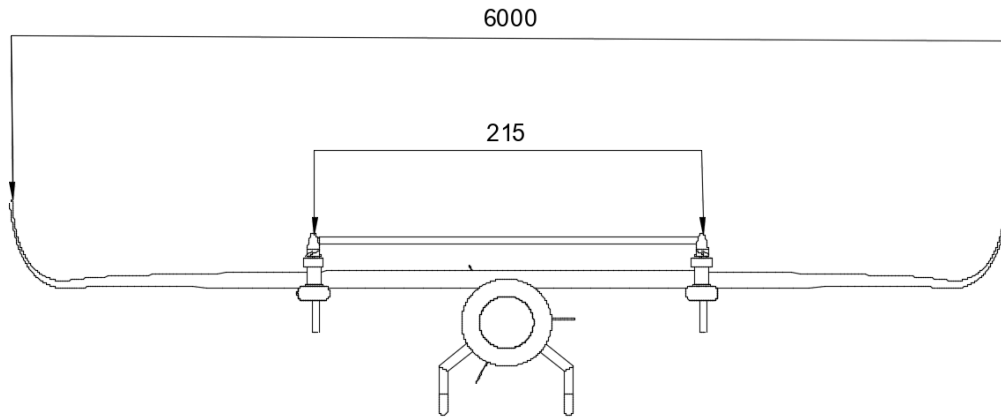


Figure 14. Front View Sketch

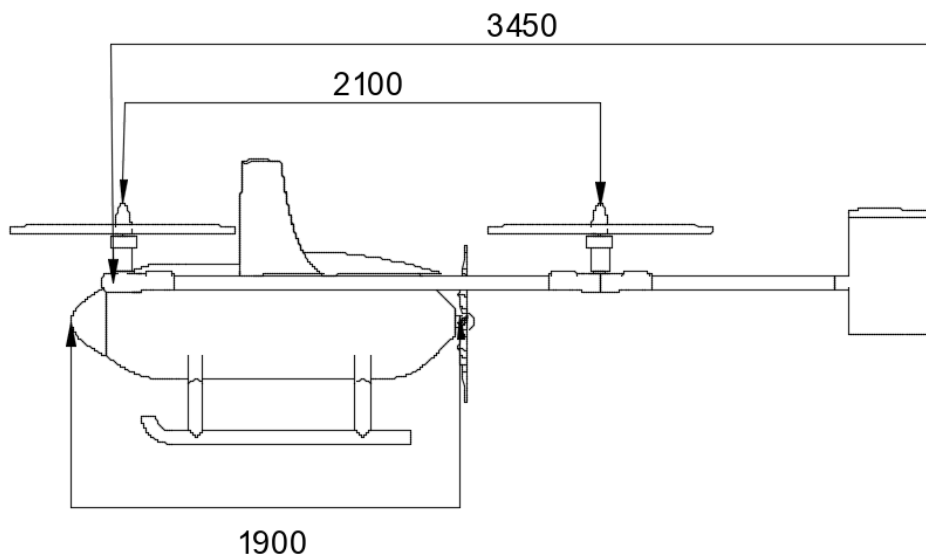


Figure 15. Side View Sketch

B WISE Equations

The heavy frames were designed based on the WISE curves, where the maximum load case for each heavy frame was resolved into tangential, radial and moment components. These were then superposed to find the total combined effect of the tangential loads (N), radial loads (N) and moment Loads (Nmm).

Tangential Load

$$M_a = \frac{PR}{2\pi} \left(\frac{3 \sin(\phi)}{2} + (\pi - \phi)(\cos(\phi) - 1) \right)$$

$$N_a = \frac{P}{2\pi} \left(\frac{\sin(\phi)}{2} + (\pi - \phi) \cos(\phi) \right)$$

$$S_a = \frac{P}{2\pi} \left((\pi - \phi) \sin(\phi) - 1 - \frac{\cos(\phi)}{2} \right)$$

Radial Unit Load

$$M_a = \frac{QR}{2\pi} \left(\frac{\cos(\phi)}{2} - (\pi - \phi) \sin(\phi) + 1 \right)$$

$$N_a = \frac{Q}{2\pi} \left(\frac{3 \cos(\phi)}{2} + (\pi - \phi) \sin(\phi) \right)$$

$$S_a = \frac{Q}{2\pi} \left((\pi - \phi) \cos(\phi) - \frac{\sin(\phi)}{2} \right)$$

Moment Unit Load

$$M_a = \frac{T}{2\pi} (\pi - 2 \sin(\phi) - \phi)$$

$$N_a = \frac{T}{2\pi R} \left(\frac{3 \cos(\phi)}{2} + (\pi - \phi) \sin(\phi) \right)$$

$$S_a = \frac{T}{2\pi R} (1 + 2 \cos(\phi))$$

C FEA Results

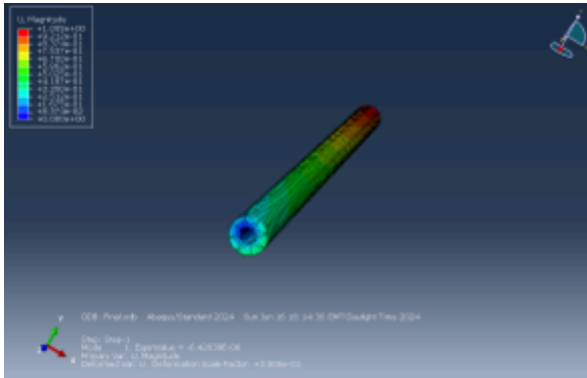


Figure 16. First Mode of Buckling

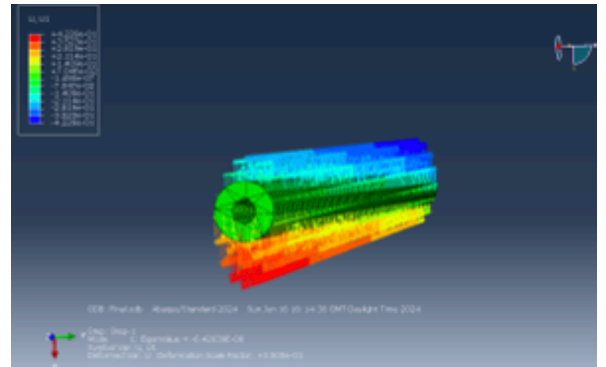


Figure 17. Total Deflection